Low-Power CMOS AI-Inspired Freshness Classifier for Biodegradable Smart Packaging

Project Report

Project Type	Cadence Virtuoso schematic, simulation, layout, DRC, LVS, and RCX verification
Prepared by	Ahmed Lutfe Inam
Technology / PDK	gpdk090, 90 nm CMOS, 1 V devices
Main Cell	ai_classifier_v2
Report Focus	Theory, circuit approach, AI-inspired weighting, simulation, layout, DRC/LVS/RCX, and power analysis

Abstract

This report presents a low-power CMOS AI-inspired freshness classifier designed as part of an EEE 4861 assignment. The project implements a simple analog hardware decision circuit for smart biodegradable food packaging. The circuit accepts three normalized sensor-related input voltages: Vgas, Vhum, and Vtemp. These inputs represent gas/spoilage risk, humidity risk, and temperature risk respectively. The output is an alarm signal that becomes high when the combined freshness risk exceeds the circuit threshold.

The term AI is used in this project in an AI-inspired hardware sense. The design is not a software neural network, not a trainable machine-learning processor, and not a deep-learning accelerator. Instead, it behaves like a very small fixed-weight classifier. The sensor voltages are weighted by using different NMOS transistor widths. A wider NMOS transistor pulls the internal decision node more strongly, so it contributes a larger weight to the decision. In this design, the gas input receives the highest weight, humidity receives a medium weight, and temperature receives the lowest weight.

The design was created in Cadence Virtuoso using gpdk090 1 V CMOS devices. The work includes schematic design, symbol/testbench creation, DC sweep simulation, transient simulation, layout implementation, DRC correction, LVS verification, RCX parasitic extraction, and power measurement. The final layout passed DRC with no errors, matched the schematic in LVS, and successfully generated an extracted view through RCX. The measured average transient supply current was -105.4 microampere over the applied 0-8 ms transient test, corresponding to 105.4 microwatt average transient power at VDD = 1 V.

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1. Introduction

Food freshness monitoring is an important application area for smart packaging. In a practical package, gas concentration, humidity, and temperature can provide useful information about the condition of stored food. For example, gas or volatile compound changes may indicate spoilage, high humidity can accelerate degradation, and high temperature can create an unsafe storage condition. A low-power circuit that combines these sensor-related signals and produces a simple alarm output can be useful for compact packaging electronics.

This project focuses on the CMOS decision circuit itself. The actual physical gas, humidity, and temperature sensors are represented by normalized voltage sources in simulation. A low voltage represents a low-risk condition, while a higher voltage represents a higher-risk condition. The purpose of the circuit is to receive these voltages and decide whether the package should be considered safe or unsafe.

The project was designed as a complete VLSI assignment flow rather than only a schematic simulation. After the schematic was created, the design was simulated, laid out manually, verified by DRC and LVS, and extracted using RCX. Therefore, the project demonstrates both circuit-level behavior and layout-level implementation steps.

Signal	**Meaning in the project**	**Low value**	**High value**
Vgas	Gas/spoilage risk sensor voltage	Low gas risk	High gas/spoilage risk
Vhum	Humidity risk sensor voltage	Low humidity risk	High humidity risk
Vtemp	Temperature risk sensor voltage	Normal temperature	High temperature risk
alarm	Digital warning output	Safe/normal	Unsafe/warning

2. Project Objective and System Idea

The main objective was to design a compact CMOS circuit that can classify the freshness condition of a package using three sensor-related inputs. The design goal was not to build a large digital processor. Instead, the goal was to implement a small analog decision circuit whose transistor sizes directly affect the decision strength of each input.

The system idea is simple: each sensor produces or is converted into a voltage between 0 V and 1 V. The circuit combines the effects of the three voltages at an internal node named nsum. The internal node is then applied to a CMOS inverter. The inverter converts the analog internal decision into a clean alarm output. If the risk is low, the alarm remains low. If the weighted risk becomes high enough, the alarm becomes high.

- Use a 1 V CMOS supply for low-voltage operation.
- Use three analog input voltages to represent environmental freshness risks.
- Give different decision importance to different sensor inputs using NMOS width ratios.
- Produce a simple alarm output that can be treated as a digital warning flag.
- Complete schematic, simulation, layout, DRC, LVS, and RCX verification in Cadence Virtuoso.

In a real packaging system, the CMOS chip would be connected to sensor interfaces. The biodegradable packaging film or substrate can be part of the package system, but the CMOS silicon circuit itself is not biodegradable. Therefore, the project is best described as a CMOS freshness classifier intended for integration with biodegradable smart packaging, not as a fully biodegradable integrated circuit.

3. Meaning of AI in This Project

The word AI is used carefully in this project. The circuit does not run Python code, does not train from data, does not update its own weights, and does not implement a deep neural network. The correct description is AI-inspired fixed-weight hardware classifier. It is inspired by the basic idea of a perceptron-like decision rule, where several inputs are multiplied by weights and compared against a threshold.

```
Risk score = wgas.Vgas + whum.Vhum + wtemp.Vtemp
```

A software AI classifier would normally calculate this score numerically. In this hardware implementation, the same idea is mapped into transistor behavior. The input voltages control NMOS gates. The widths of the NMOS transistors define how strongly each input can pull down the decision node. Therefore, the circuit uses physical transistor sizing as a fixed hardware weight.

```
if Risk score > Threshold, then alarm = 1
```

The threshold is not stored as a digital number. It is created by the ratio between the PMOS load pull-up strength, the NMOS pull-down strengths, and the switching threshold of the output inverter. This is why the circuit is compact: it performs a simple classification action directly through analog CMOS device behavior.

AI/ML idea	Hardware equivalent in this project
Input features	Vgas, Vhum, and Vtemp analog voltages
Weights	NMOS transistor widths W = 4 um, 2 um, and 1 um
Weighted summation	Combined pull-down action at internal node nsum
Threshold decision	Balance of PMOS load and NMOS network, followed by inverter threshold
Classifier output	alarm signal
Training	Not implemented; weights are fixed by design

4. CMOS Theory Used in the Classifier

4.1 NMOS Pull-Down Strength

An NMOS transistor conducts more strongly when its gate voltage increases above threshold. In this project, each sensor voltage is connected to the gate of an NMOS transistor. When the sensor voltage is low, the NMOS is weak or off. When the sensor voltage becomes high, the NMOS turns on and tries to pull the internal node nsum toward ground.

For the same technology and channel length, the current-driving strength of a MOS transistor approximately increases with its channel width W. A wider transistor has a larger channel area, so it can conduct more current for the same gate voltage. This is the physical reason why width can be used as a circuit-level weight.

```
Larger W  ->  larger pull-down current  ->  stronger input weight
```

The MOS square-law equation is only an approximation, especially in short-channel CMOS processes, but it is useful for understanding the design. If other parameters are fixed, increasing W increases the drain current. Therefore, the gas NMOS with W = 4 um contributes more strongly than the humidity NMOS with W = 2 um, and much more strongly than the temperature NMOS with W = 1 um.

4.2 PMOS Load and Internal Node

The PMOS load transistor MLOAD is connected as an always-on pull-up device. Its source is connected to VDD and its gate is tied to ground, so it continuously tries to pull nsum upward. The NMOS sensor transistors are connected as pull-down devices. Therefore, nsum becomes the competition node between the pull-up load and the weighted pull-down network.

If sensor risks are low, the NMOS pull-down network is weak. MLOAD keeps nsum high. The output inverter sees a high input and produces a low alarm. If the sensor risks increase, the NMOS network becomes stronger. Then nsum is pulled down. The inverter sees a low input and produces a high alarm. This gives the desired safe-to-warning transition.

4.3 Output Inverter

The internal node nsum is not directly used as the final output because it is an analog decision node. A CMOS inverter is used after nsum to regenerate a clean output. The inverter consists of MPINV and MNINV. When nsum is high, the output alarm is low. When nsum is low, the output alarm is high. This inversion is useful because a strong pull-down at nsum indicates high risk, and the final alarm should become high during high risk.

5. Proposed Circuit Architecture

The complete classifier circuit contains six MOS transistors. One PMOS transistor is used as the load, three NMOS transistors are used as weighted sensor-controlled pull-down paths, and two transistors form the output inverter. The circuit is intentionally simple so that the weighting and decision mechanism can be understood from transistor-level behavior.

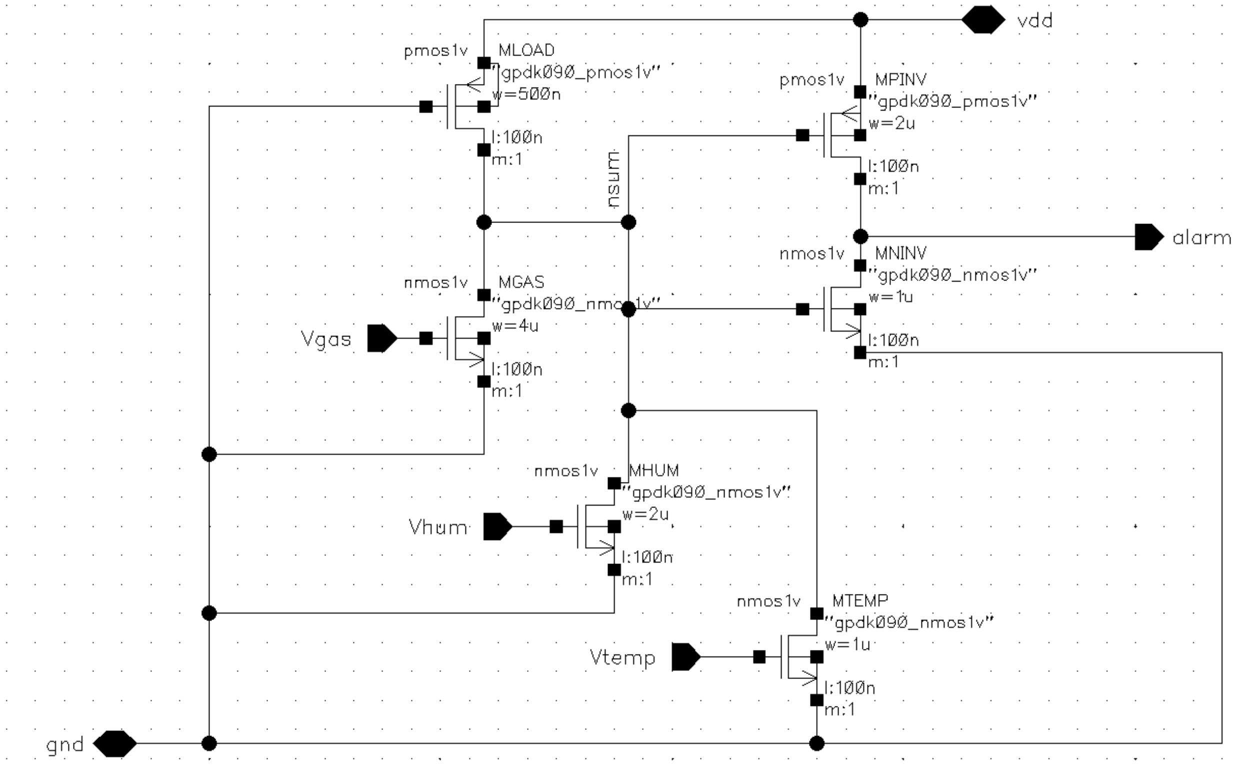

Figure 1: Schematic of the proposed CMOS AI-inspired freshness classifier. Vgas, Vhum, and Vtemp control weighted NMOS pull-down devices; nsum is the internal decision node; alarm is generated by the output inverter.

The schematic shows that MLOAD pulls nsum upward, while MGAS, MHUM, and MTEMP pull nsum downward depending on their gate voltages. The right side of the schematic is a CMOS inverter. The input of the inverter is nsum, and its output is alarm. This makes the alarm signal high when nsum becomes sufficiently low.

Transistor	Type	Role	Width W	Length L
MLOAD	PMOS	Always-on pull-up load for nsum	500 nm	100 nm
MGAS	NMOS	Gas input weighted pull-down	4 um	100 nm
MHUM	NMOS	Humidity input weighted pull-down	2 um	100 nm
MTEMP	NMOS	Temperature input weighted pull-down	1 um	100 nm
MPINV	PMOS	Pull-up device of output inverter	2 um	100 nm
MNINV	NMOS	Pull-down device of output inverter	1 um	100 nm

6. Device Sizing and Weight Selection

The most important design idea is the use of transistor width as the hardware weight. The channel length of the three sensor NMOS transistors is kept the same at 100 nm. Their widths are different: MGAS is 4 um, MHUM is 2 um, and MTEMP is 1 um. Therefore, the approximate weight ratio is 4:2:1.

```
Weight ratio = Wgas : Whum : Wtemp = 4 um : 2 um : 1 um = 4 : 2 : 1
```

This sizing means that gas is treated as the most important freshness indicator in the classifier. A smaller gas voltage can trigger the alarm compared with humidity and temperature because MGAS can pull down nsum more strongly. Humidity has medium importance, and temperature has the weakest single-input effect.

Input	NMOS device	W/L	Relative weight	Expected effect
Vgas	MGAS	4 um / 100 nm	4	Strongest pull-down; lowest switching input among the three
Vhum	MHUM	2 um / 100 nm	2	Medium pull-down; medium switching input
Vtemp	MTEMP	1 um / 100 nm	1	Weakest pull-down; highest switching input

The PMOS load width was selected as 500 nm. This makes the pull-up weak enough that the weighted NMOS network can pull nsum down when risk increases, but still strong enough to keep nsum high when the inputs are low. The output inverter uses MPINV = 2 um and MNINV = 1 um, giving a conventional PMOS-to-NMOS width ratio that helps produce a clean output transition.

This circuit is therefore a ratioed logic structure. Its switching behavior depends on the ratio of the PMOS load strength, the NMOS pull-down strengths, and the inverter threshold. Because it is ratioed, it can consume static current when the pull-up and pull-down paths are both conducting. This is why power analysis is an important part of the report.

7. Cadence Implementation Flow

The design was implemented in Cadence Virtuoso using gpdk090 CMOS devices. The main schematic cell was named ai_classifier_v2. After the transistor-level schematic was completed, a symbol was created and placed in a testbench for simulation. The testbench contains DC voltage sources for Vgas, Vhum, Vtemp, and VDD. For transient simulation, the Vgas input was also applied as a pulse/ramp waveform to observe alarm switching with time.

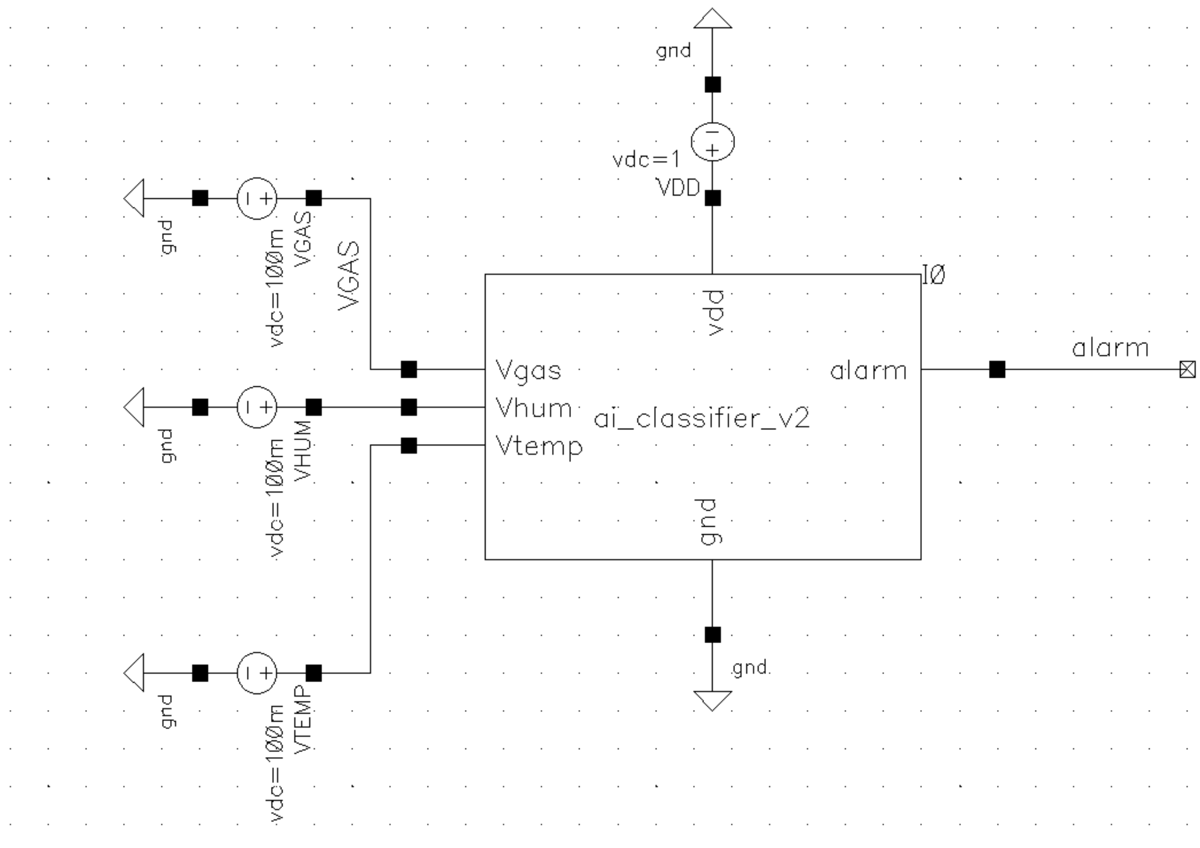

Figure 2: Testbench used for the classifier simulation. The classifier symbol receives Vgas, Vhum, Vtemp, VDD, and GND connections, and the alarm output is monitored.

The testbench confirms the modular nature of the design. The inner transistor circuit is wrapped as a symbol, and the symbol is stimulated by external sensor-voltage sources. This is similar to a real system where the classifier block would receive sensor interface voltages from packaging sensors.

7.1 Design Steps Followed

1. Created a transistor-level schematic using gpdk090 NMOS and PMOS 1 V devices.
2. Named the important nets: Vgas, Vhum, Vtemp, vdd, gnd, nsum, and alarm.
3. Sized the three sensor NMOS transistors with widths 4 um, 2 um, and 1 um to realize fixed hardware weights.
4. Added the PMOS load and the output inverter.
5. Created a symbol for the classifier cell.
6. Built a testbench using voltage sources for input and supply signals.
7. Ran DC sweeps to find switching behavior for each input.
8. Ran transient simulation to show time-domain alarm response.
9. Implemented the layout manually.
10. Fixed DRC errors, verified LVS matching, and generated the extracted view using RCX.

8. Simulation Setup and Results

Two main types of simulation were performed: DC sweep and transient analysis. In the DC sweep, one sensor input was swept from 0 V to 1 V while the other inputs were kept at a low value. This shows how strongly each input can trigger the alarm. In the transient analysis, Vgas was changed with time to observe how the alarm responds dynamically.

8.1 DC Sweep of Vgas

The Vgas sweep verifies the strongest weighted input. Since MGAS has the largest width of 4 um, it is expected to turn the alarm on at a lower input voltage than the other inputs. The simulation confirms this behavior. The alarm begins switching around approximately 0.35 V, showing that the gas path dominates the decision when it rises.

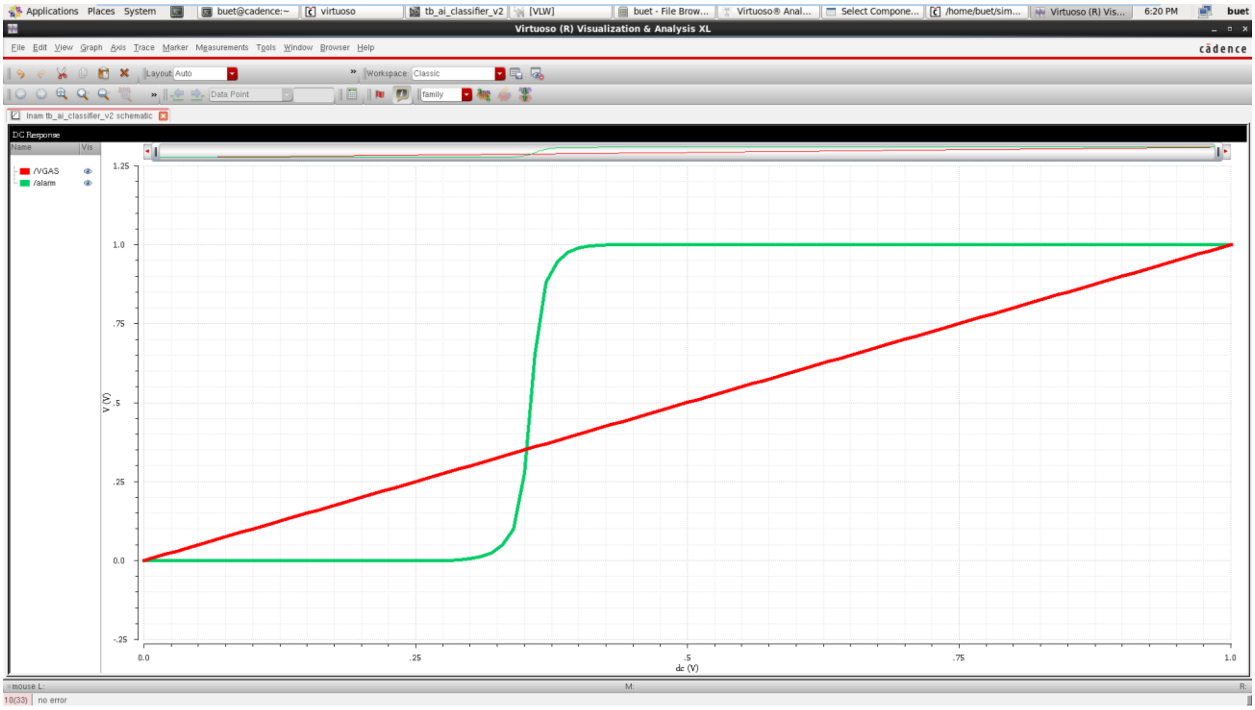

Figure 3: DC sweep of Vgas versus alarm output. Vgas is the strongest weighted input because MGAS has the largest width.

8.2 DC Sweep of Vhum

The Vhum sweep shows the medium-weight behavior. The humidity input is connected to MHUM with W = 2 um. Since this width is smaller than MGAS but larger than MTEMP, the switching point appears after the gas input case but before the temperature input case. The alarm switches around approximately 0.40-0.42 V.

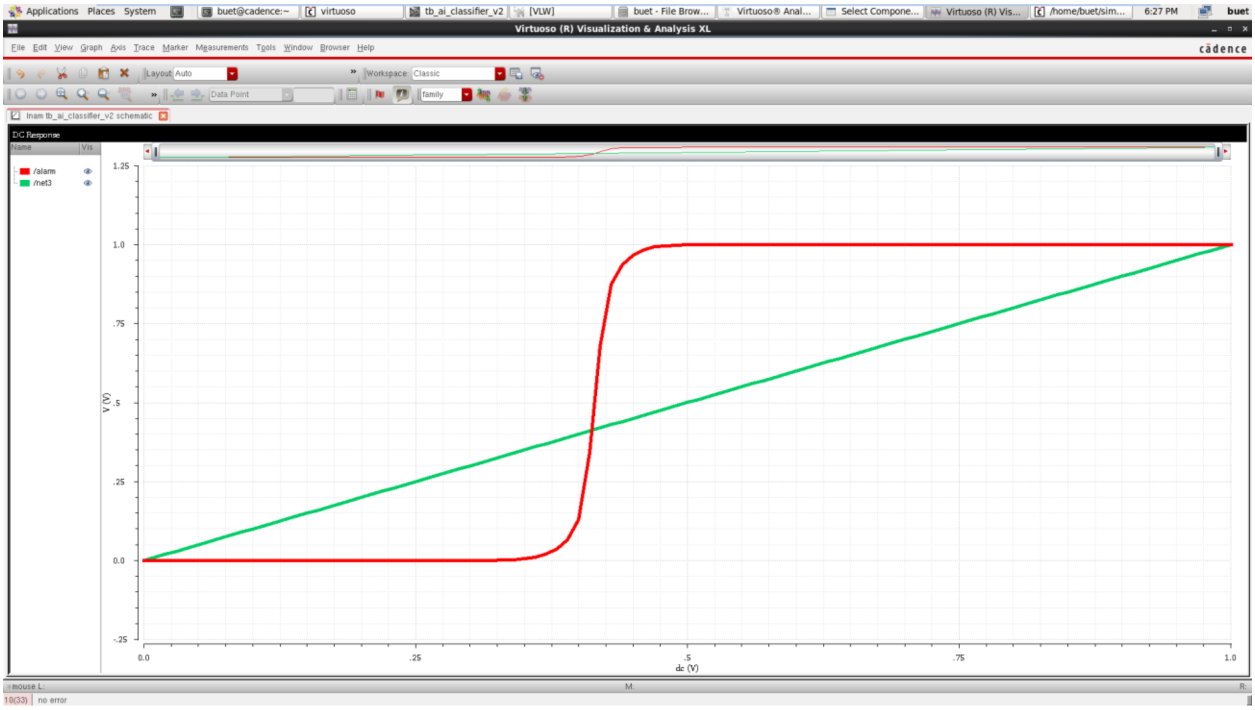

Figure 4: DC sweep of Vhum versus alarm output. The humidity input has medium decision weight due to the 2 um NMOS width.

8.3 DC Sweep of Vtemp

The Vtemp sweep shows the weakest single-input behavior. MTEMP has W = 1 um, so it produces the weakest pull-down current among the three sensor-controlled NMOS devices. Therefore, the temperature input needs a higher voltage to make the alarm switch. The plot shows switching near approximately 0.50 V.

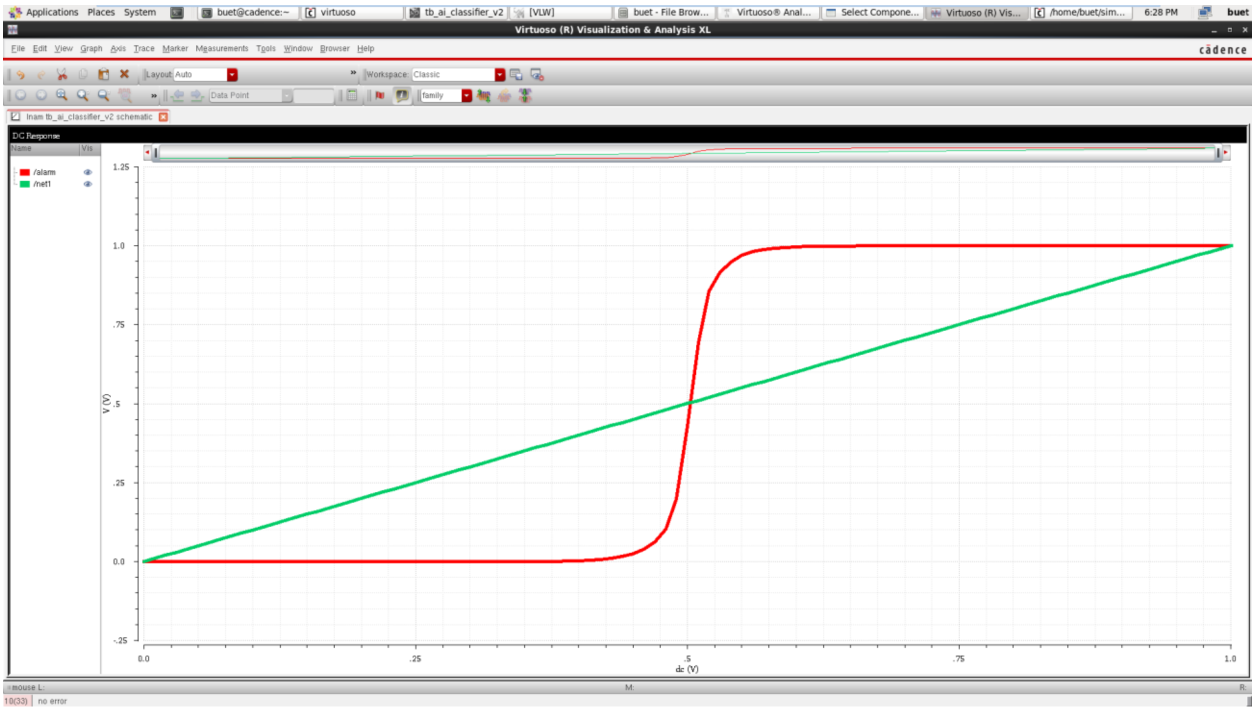

Figure 5: DC sweep of Vtemp versus alarm output. The temperature input has the weakest single-input decision weight due to the 1 um NMOS width.

The three DC sweeps prove the hardware weighting concept. A larger NMOS width gives a stronger pull-down, and a stronger pull-down causes the alarm to switch at a lower input voltage. Therefore, the ordering of the switching behavior is consistent with the designed weight ratio: gas first, humidity second, and temperature third.

Input swept	**Device width**	**Approximate alarm switching voltage**	**Interpretation**
Vgas	4 um	~0.35 V	Strongest weight; switches earliest
Vhum	2 um	~0.40-0.42 V	Medium weight
Vtemp	1 um	~0.50 V	Weakest weight; switches latest

8.4 Transient Response

For transient simulation, Vgas was applied as a time-varying input. It started at a safe low value, increased toward a high-risk value, remained high for a period, and then returned low. The alarm waveform follows the expected classifier behavior: the alarm is initially low, becomes high when Vgas crosses the decision region, and returns low after Vgas falls back below the threshold.

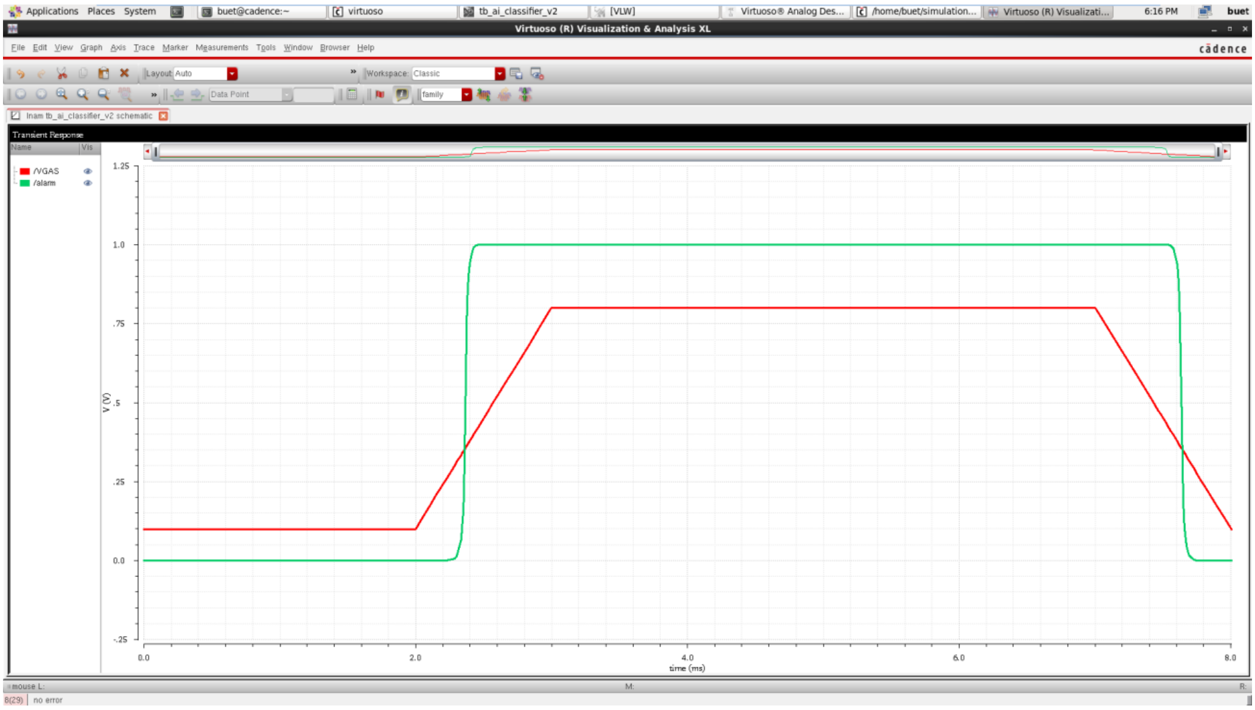

Figure 6: Transient response of Vgas and alarm. The alarm turns on when the weighted gas input makes the internal decision node cross the inverter threshold.

This transient result is important because it shows that the circuit is not only a DC transfer characteristic. It can respond to a changing sensor input in time. The alarm is a real-time output, not a latch. Therefore, when the risk input decreases again, the output can return to the safe state. A latch could be added in future work if a sticky alarm is required.

9. Power Consumption Analysis

Power was measured by observing the current through the VDD voltage source during transient simulation. In Cadence, the supply current waveform appeared negative because of the current direction convention of the voltage source. This does not mean negative power consumption. It means that the voltage source is delivering current to the circuit. The magnitude of the current is used for consumed power.

$$P(t) = VDD \times |IDD(t)|$$

Since the supply voltage is 1 V, the numerical value of current in microampere is equal to the numerical value of power in microwatt. For example, 105.4 microampere at 1 V corresponds to 105.4 microwatt.

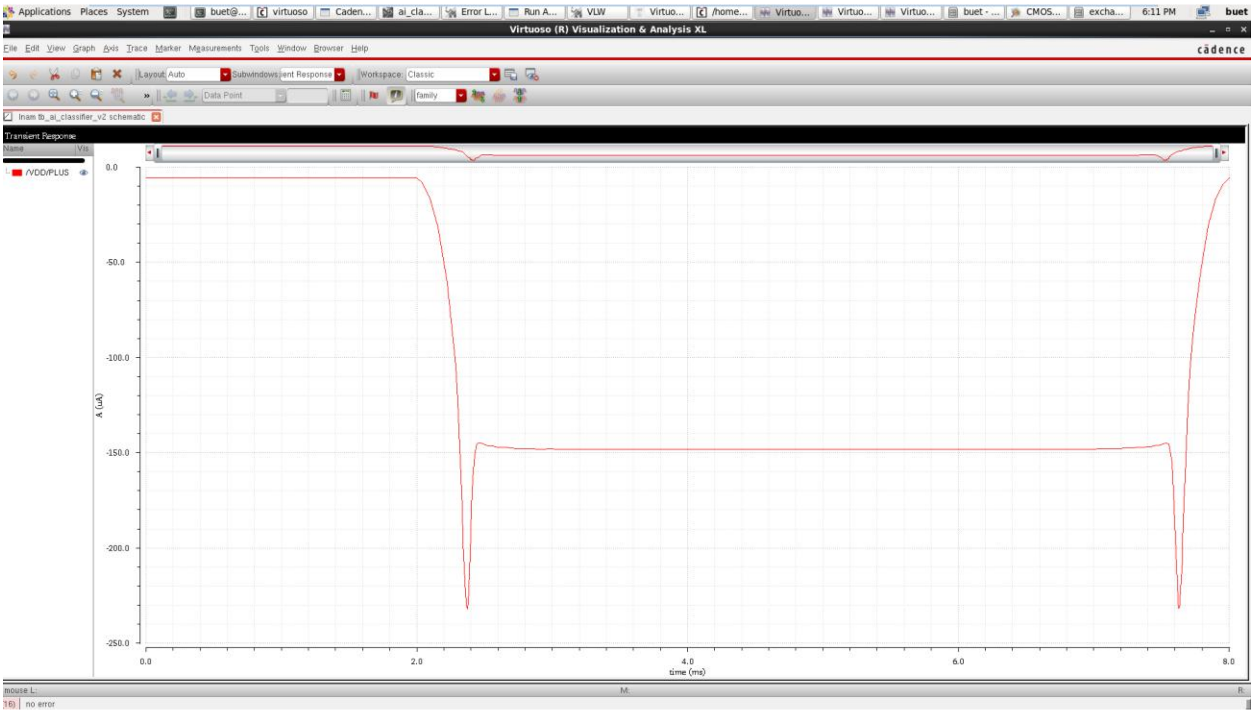

Figure 7: Transient VDD supply current waveform. The negative sign is due to the Cadence current direction convention. The magnitude is used for power calculation.

From the waveform, the circuit shows a low idle current when the circuit is in the safe state, a higher current when the alarm/high-risk state is active, and a larger short switching peak during transitions. The static current in the alarm condition occurs because this is a ratioed circuit: the PMOS load and the NMOS pull-down network can conduct at the same time.

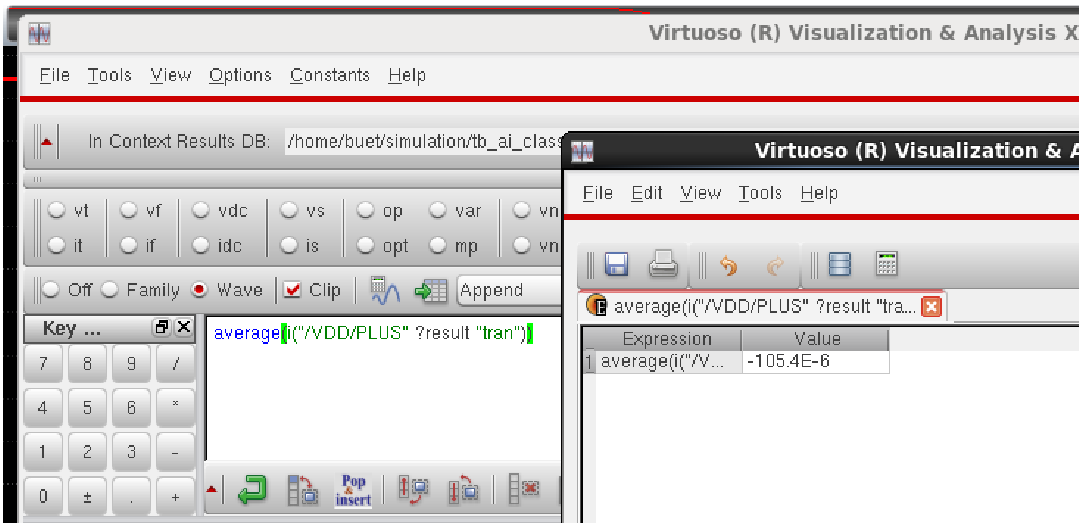

Figure 8: Cadence Calculator result showing average VDD current of approximately -105.4 microampere for the transient simulation interval.

The average current measured from the Cadence Calculator was -105.4 microampere. Taking the magnitude gives 105.4 microampere. Since VDD = 1 V, the average transient power is 105.4 microwatt for the applied 0-8 ms input pattern.

```
Pavg = VDD x |Iavg| = 1 V x 105.4 uA = 105.4 uW

E(0 to 8 ms) = Pavg x T = 105.4 uW x 8 ms = 0.843 uJ
```

Power metric	Approximate value	Meaning
Idle/safe-state power	~5-10 uW	Low-risk input condition, alarm low
Alarm-state static power	~145-150 uW	High-risk condition, alarm high
Peak switching power	~230 uW	Short transition spike during switching
Average transient power	105.4 uW	Average over the simulated 0-8 ms transient input pattern
Energy for 0-8 ms test	~0.843 uJ	Total energy used during the simulated transient interval

The value 105.4 microwatt should be written as average transient power over the simulated 0-8 ms test condition. It is not a universal lifetime power value for all possible sensor conditions. If the circuit remains mostly in the safe state, its long-term average can be much lower. If it remains in the alarm state, the average can be closer to the alarm-state static power. Therefore, the long-term packaging-system power depends on duty cycle and operating condition.

10. Layout Design Approach

After schematic simulation, the layout was created manually in Cadence Virtuoso. The layout follows a simple standard-cell-like placement style. The VDD rail is placed at the top, and the GND rail is placed at the bottom. PMOS devices are placed near the VDD rail, while NMOS devices are placed closer to the GND rail. This arrangement simplifies power routing, substrate/body connections, and signal routing.

The three weighted NMOS devices were laid out with different widths. The gas transistor is physically the widest, the humidity transistor is medium-sized, and the temperature transistor is smaller. This visual layout difference directly represents the 4:2:1 weighting concept used in the schematic.

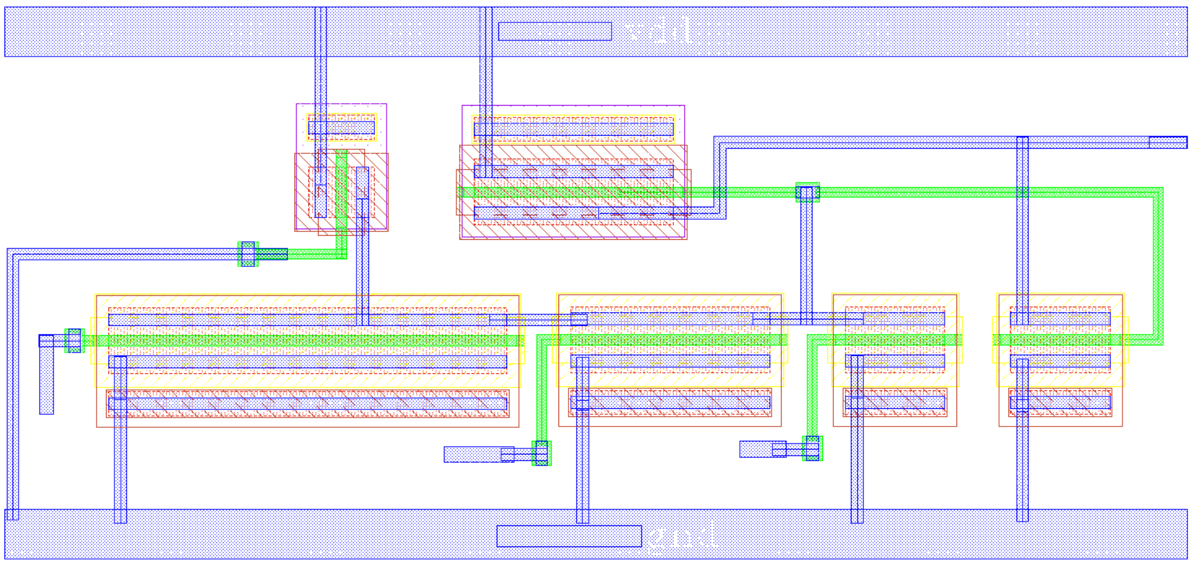

Figure 9: White-background colored layout view of the classifier. PMOS devices are placed near VDD and NMOS devices are placed near GND, with input and output routing visible.

The layout includes body connections to reduce body-effect and substrate problems. PMOS bodies/n-wells are tied to VDD, and NMOS bodies/substrate taps are tied to GND. The main pins exported in the layout are vdd, gnd, Vgas, Vhum, Vtemp, and alarm. The internal decision node nsum is routed inside the layout and is not treated as an external pin.

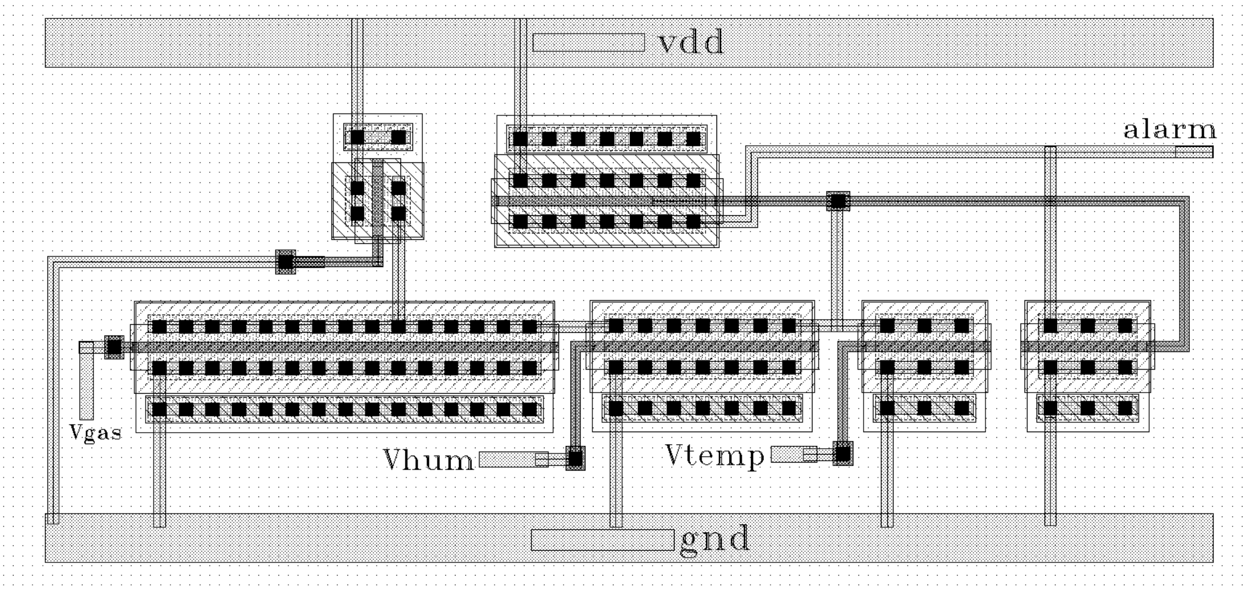

Figure 10: Print-friendly white-background layout view with visible device internals and labels.

Layout item	Implementation detail
Power rails	VDD rail at top, GND rail at bottom
PMOS placement	MLOAD and MPINV placed near VDD rail inside n-well region
NMOS placement	MGAS, MHUM, MTEMP, and MNINV placed near GND rail
Input pins	Vgas, Vhum, and Vtemp routed to corresponding NMOS gates
Output pin	alarm routed from the inverter output
Internal node	nsum connects PMOS load, NMOS pull-down network, and inverter input
Body contacts	PMOS body tied to VDD; NMOS body/substrate tied to GND

11. DRC, LVS, RCX, and Extracted View

11.1 DRC Verification

Design Rule Check verifies whether the layout follows the physical manufacturing rules of the selected technology. The first DRC run did not pass. The initial errors included n-well spacing, implant spacing, and metal width related issues. The layout was then corrected by increasing spacing between well/implant regions and widening the metal segment that violated the minimum metal width rule.

Initial DRC issue	Meaning	Fix applied
N-well spacing violation	Same-potential n-well regions were too close	Increased spacing between PMOS/n-well regions
P-implant spacing violation	P-implant regions were too close	Adjusted spacing around implant/tap regions
Metal1 width violation	A Metal1 segment was below minimum width	Widened the Metal1 route

After fixing these issues, DRC was run again and the layout passed with no DRC errors. This means the final layout satisfies the physical design rules checked by the Assura DRC setup.

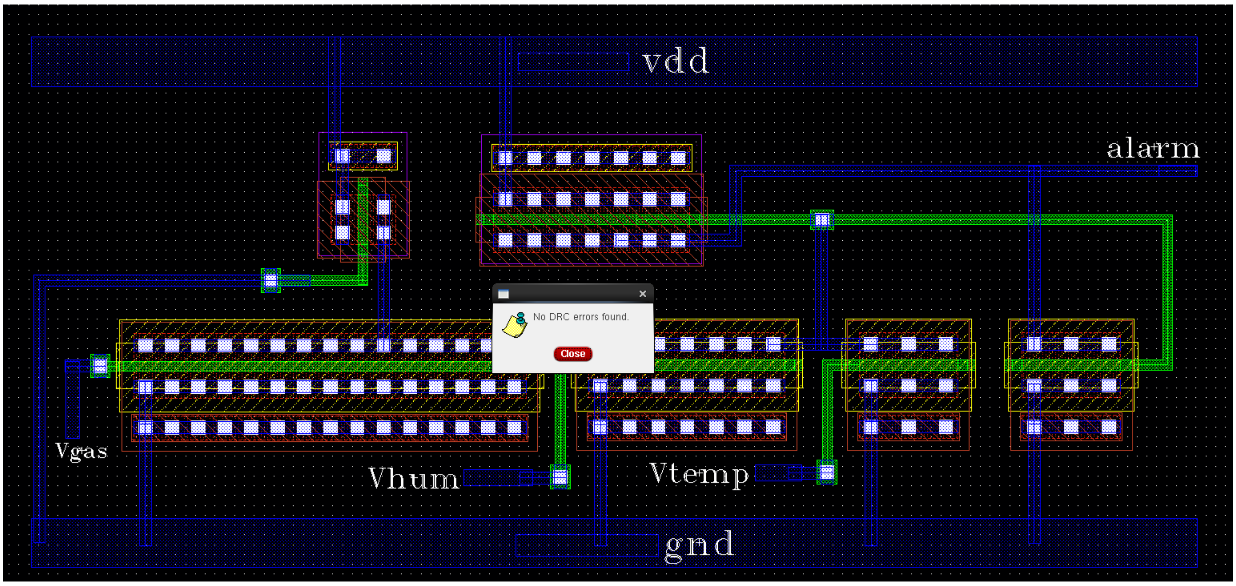

Figure 11: Final DRC result showing no DRC errors after the initial layout violations were corrected.

11.2 LVS Verification

Layout Versus Schematic verifies whether the drawn layout electrically matches the original schematic. This step is critical because a layout can pass DRC but still be electrically wrong. LVS compares device types, device counts, pins, nets, and parameters between the schematic and the extracted layout representation.

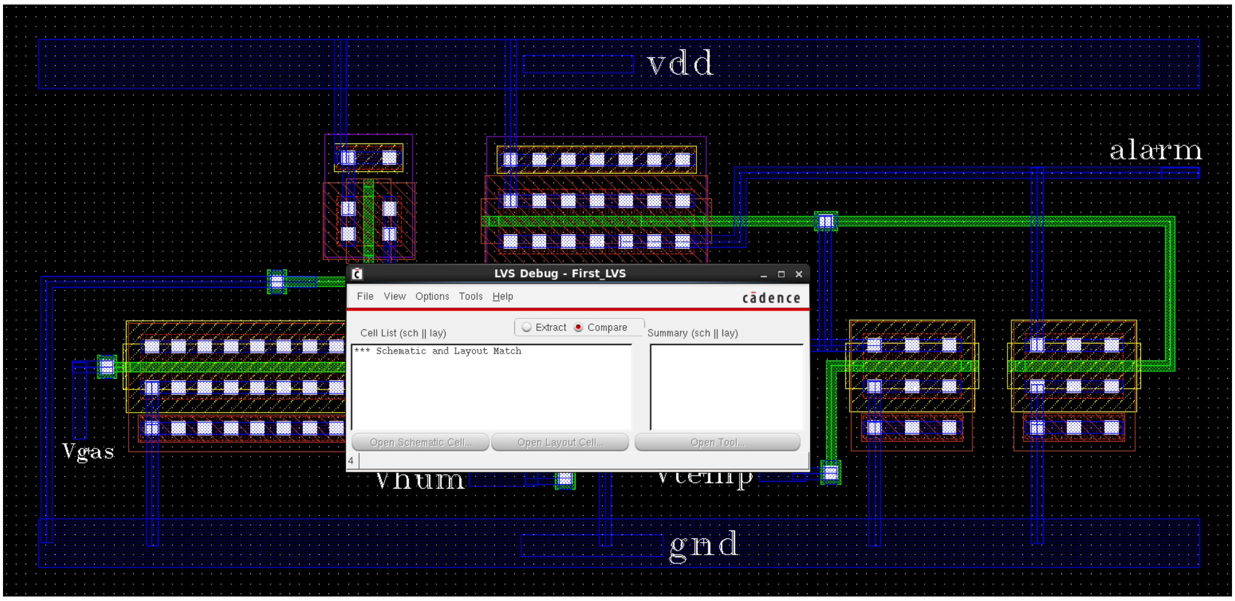

Figure 12: LVS result showing that schematic and layout match.

The LVS summary shows zero net mismatches, zero device mismatches, zero pin mismatches, and zero parameter mismatches. This confirms that the layout is not only physically clean but also electrically equivalent to the schematic design.

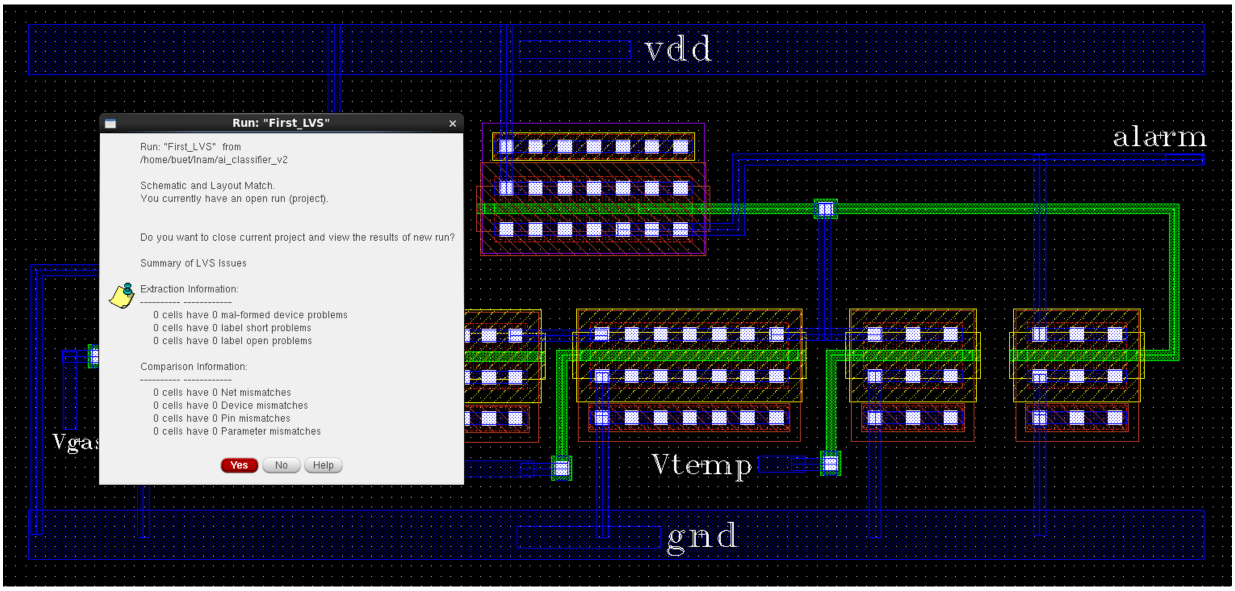

Figure 13: LVS debug/details window confirming schematic and layout match.

11.3 RCX Parasitic Extraction

After DRC and LVS, RCX parasitic extraction was performed. RCX extracts parasitic resistance and capacitance from the layout routing and creates an extracted view. This extracted view is important because real layout wires and contacts introduce parasitic effects that are not visible in the ideal schematic.

In the RCX setup, the extraction output was generated as an av_extracted view. The extraction mode used decoupled capacitance, full-chip net selection, and the gpdk090 technology setup. The RCX run completed successfully, confirming that an extracted representation of the layout was generated.

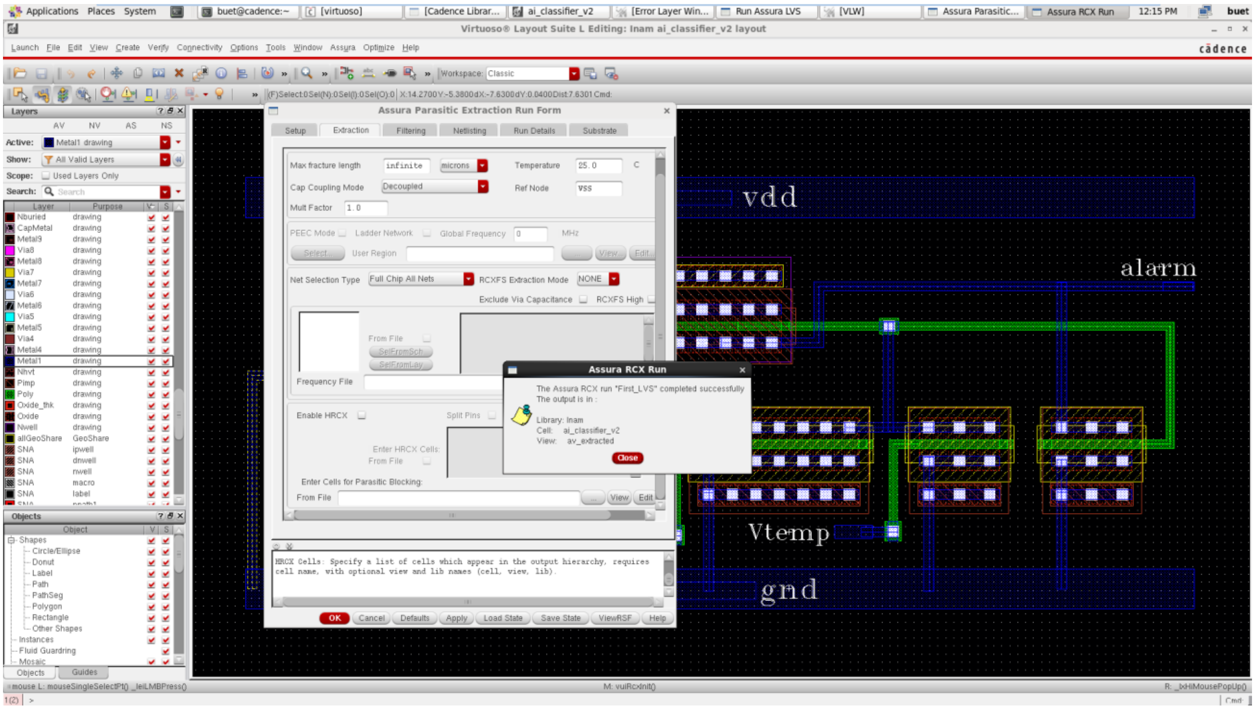

Figure 14: Assura RCX run completed successfully and generated the av_extracted view.

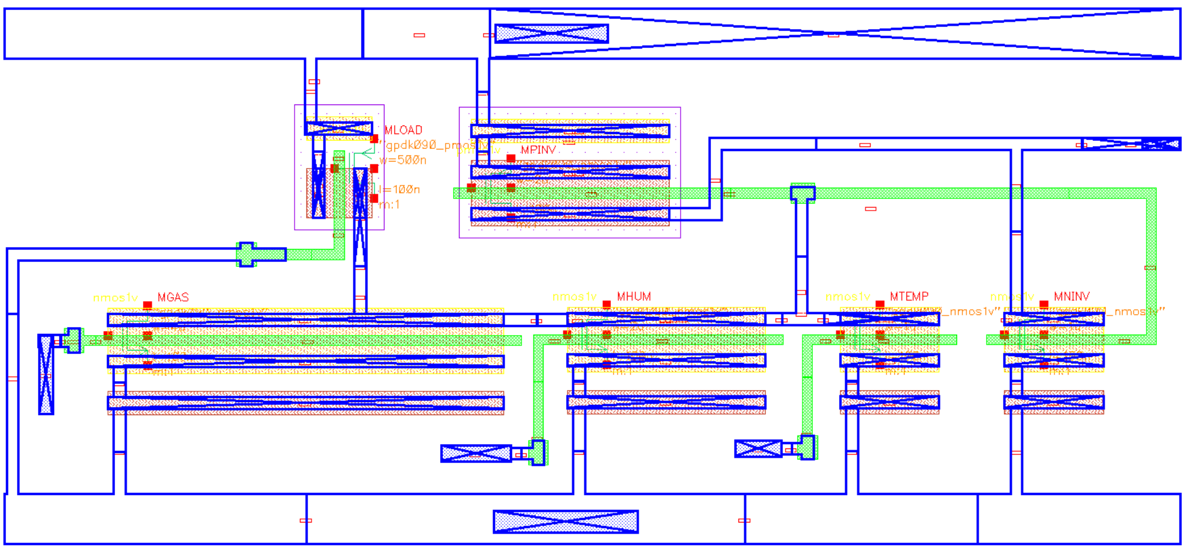

Figure 15: Generated av_extracted view after RCX parasitic extraction. This view contains extracted layout information and parasitic-related representation.

The av_extracted view can look visually different from the clean layout because it displays extracted devices, parasitic annotations, and net information. It should not be judged as a clean layout picture. Its purpose is verification and post-layout simulation. For report presentation, the clean layout image is better for visual explanation, while the av_extracted image is useful as proof of successful extraction.

12. Discussion, Limitations, and Future Work

12.1 What the Project Successfully Demonstrates

- A compact six-transistor CMOS classifier circuit was designed.
- The AI-inspired weighted decision idea was mapped into transistor widths.
- DC simulation confirmed that different transistor widths produce different switching sensitivity.
- Transient simulation confirmed real-time alarm switching behavior.
- Layout was completed and corrected after initial DRC errors.

- Final DRC passed with no errors.
- LVS confirmed that the layout matches the schematic.
- RCX successfully generated an extracted view.
- Power was measured using VDD supply current and reported in the microwatt range.

12.2 Limitations

The circuit is intentionally simple and fixed-weight. It does not learn from data, cannot automatically tune its threshold, and cannot adapt to different food types without redesign or calibration. The sensor inputs were modeled as voltage sources, so the actual sensor interface circuits were outside the scope of this assignment. Also, because the circuit uses a ratioed PMOS load structure, it can draw static current when the NMOS pull-down network is active.

The average power value of 105.4 microwatt was calculated for the simulated 0-8 ms transient input pattern. A different input duty cycle will produce a different long-term average power. For example, if the circuit spends most of its time in the safe state, the average power will be closer to the idle power. If it spends most of its time in alarm state, the average power will be closer to the alarm-state power.

12.3 Future Work

- Run post-layout simulations using the av_extracted view to compare schematic and parasitic-aware behavior.
- Test process, voltage, and temperature variations for more robust operation.
- Add an ultra-low-power latch if the alarm needs to remain stored after a spoilage event.
- Design sensor interface blocks for gas, humidity, and temperature sensors.
- Use a current-source or more controlled load to reduce static power compared with the simple PMOS-load structure.
- Add calibration switches or programmable transistor arrays to tune the effective weights.
- Design a pad ring and ESD protection if the project is extended toward a chip-level implementation.

13. Conclusion

This project demonstrates a compact low-power CMOS AI-inspired freshness classifier for biodegradable smart packaging applications. The circuit uses three sensor-related analog input voltages and produces a digital alarm output. The AI-inspired behavior comes from the fixed-weight classifier concept, where transistor widths implement different weights for gas, humidity, and temperature inputs.

The design uses a 4:2:1 NMOS width ratio, which gives gas the highest decision weight, humidity a medium weight, and temperature the lowest weight. DC sweep simulation confirms this: Vgas switches the alarm at the lowest input voltage, followed by Vhum and then Vtemp. Transient simulation confirms that the alarm turns on when the input risk rises and turns off when the risk decreases.

The project also completes important VLSI verification steps. The layout was created manually, initial DRC issues were fixed, the final DRC passed with no errors, LVS confirmed schematic-layout matching, and RCX generated the av_extracted view. Power analysis showed idle power around 5-10 microwatt, alarm-state power around 145-150 microwatt, peak switching power around 230 microwatt, and average transient power of 105.4 microwatt for the 0-8 ms input test. Overall, the project is a layout-verified and fabrication-oriented CMOS classifier demonstration suitable for an EEE 4861 VLSI assignment.

Appendix A: Report-Ready Summary

The proposed design is a fixed-weight CMOS classifier that maps three environmental freshness indicators into a single alarm output. The gas, humidity, and temperature inputs are treated as analog voltages. The weighted decision is implemented physically by using different NMOS widths. Wider NMOS devices pull the decision node down more strongly and therefore represent higher weights. The PMOS load provides the pull-up action, and the inverter converts the internal analog decision into a digital alarm output.

Item	**Final project result**
Technology	gpdk090 90 nm CMOS, 1 V devices
Circuit size	6 MOS transistors
Sensor inputs	Vgas, Vhum, Vtemp
Decision node	nsum
Output	alarm
Weight implementation	NMOS width ratio 4:2:1
DRC	Passed after initial violations were corrected
LVS	Schematic and layout matched
RCX	Completed successfully; av_extracted view generated
Average transient power	105.4 uW over 0-8 ms input test